Chapter 17

Locomotion and Movement

Movement is one of the significant features of living beings, and animals and plants show a wide range of it.

From streaming protoplasm to limbs

- the streaming of protoplasm in unicellular organisms such as Amoeba is a simple form of movement
- many organisms show movement of cilia, flagella and tentacles
- human beings can move limbs, jaws, eyelids, tongue and more
- some of these movements bring a change of place or location, and such voluntary movements are called locomotion, as in walking, running, climbing, flying and swimming

Why the two cannot be separated

- locomotory structures need not differ from those used for other kinds of movement
- in Paramoecium, cilia move food through the cytopharynx and also carry out locomotion, Hydra uses its tentacles both to capture prey and to move, and we use our limbs both to change posture and to travel
- so all locomotions are movements, but all movements are not locomotions

What locomotion is for

- the methods animals use vary with their habitats and with what the situation demands
- locomotion is generally for the search of food, shelter, a mate, suitable breeding grounds or favourable climatic conditions, or to escape from enemies and predators

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17.1 Types of movement

Amoeboid and ciliary

- cells of the human body show three main types of movement, amoeboid, ciliary and muscular
- specialised cells such as macrophages and the leucocytes of blood show amoeboid movement, brought about by pseudopodia formed by the streaming of protoplasm as in Amoeba, with cytoskeletal elements such as microfilaments also involved
- ciliary movement occurs in most of our internal tubular organs, which are lined by ciliated epithelium
- the coordinated movement of cilia in the trachea helps remove dust particles and some of the foreign substances inhaled with the air, and the passage of ova through the female reproductive tract is also helped by ciliary movement

Muscular

- movement of our limbs, jaws, tongue and the like calls for muscular movement
- the contractile property of muscles is used effectively for locomotion and other movements by human beings and by most multicellular organisms
- locomotion needs a perfectly coordinated activity of the muscular, skeletal and neural systems
- flagellar movement, by contrast, helps the swimming of spermatozoa, the maintenance of the water current in the canal system of sponges and the locomotion of Protists such as Euglena, cilia and flagella being outgrowths of the cell membrane

17.2 Muscle

A tissue with four properties

- muscle is a specialised tissue of mesodermal origin
- about 40-50 per cent of the body weight of a human adult is contributed by muscles
- they have the special properties of excitability, contractility, extensibility and elasticity
- muscles have been classified by location, by appearance and by the nature of the regulation of their activities, and by location three types are identified, skeletal, visceral and cardiac

Skeletal and visceral muscle

- skeletal muscles are closely associated with the skeletal components of the body, look striped under the microscope and so are called striated, and since their activity is under the voluntary control of the nervous system they are voluntary muscles, involved primarily in locomotory actions and changes of body posture
- visceral muscles lie in the inner walls of hollow visceral organs such as the alimentary canal and the reproductive tract, show no striation and are smooth in appearance, hence smooth or nonstriated muscle
- their activity is not under the voluntary control of the nervous system, so they are involuntary, and they assist in moving food through the digestive tract and gametes through the genital tract

Fascicle, the muscle bundle

Muscle fibre, the muscle cell within the bundle

Sarcolemma, the lining of each fibre

Blood capillary, running among the fibres

Figure 17.A Diagrammatic cross sectional view of a muscle showing muscle bundles and muscle fibres

17.3 The skeletal muscle

Cardiac muscle

- cardiac muscles, as the name suggests, are the muscles of the heart
- many cardiac muscle cells assemble in a branching pattern to form a cardiac muscle
- by appearance they are striated, and they are involuntary, since the nervous system does not control their activity directly

Bundles, fibres and sarcoplasm

- each organised skeletal muscle is made of a number of muscle bundles or fascicles held together by a common collagenous connective tissue layer called fascia, and each bundle holds a number of muscle fibres
- each fibre is lined by a plasma membrane, the sarcolemma, enclosing the sarcoplasm
- a muscle fibre is a syncitium, since the sarcoplasm contains many nuclei
- the endoplasmic reticulum of the fibre, that is the sarcoplasmic reticulum, is the store house of calcium ions

Myofibrils and their bands

- a characteristic feature of the muscle fibre is a large number of parallelly arranged filaments in the sarcoplasm, the myofilaments or myofibrils
- each myofibril carries alternate dark and light bands
- the striated appearance comes from the distribution pattern of two important proteins, actin and myosin
- the light bands hold actin and are called the I-band or isotropic band, while the dark band, the A or anisotropic band, holds myosin

A band, the dark band of thick myosin filaments

I band, the light band of thin actin filaments

Z line, bisecting each I band

M line, holding the thick filaments in the middle of the A band

H zone, the part of the thick filament left free at rest

Sarcomere, the stretch between two successive Z lines

Figure 17.B Diagrammatic representation of (a) anatomy of a muscle fibre showing a sarcomere (b) a sarcomere

17.4 Bands and the sarcomere

Thin and thick filaments

- both proteins are arranged as rod-like structures, parallel to each other and to the longitudinal axis of the myofibrils
- actin filaments are thinner than the myosin filaments, so the two are commonly called the thin and the thick filaments

The lines that anchor them

- in the centre of each I band is an elastic fibre, the Z line, which bisects it, and the thin filaments are firmly attached to it
- the thick filaments in the A band are held together in the middle of that band by a thin fibrous membrane, the M line
- the A and I bands are arranged alternately throughout the length of the myofibrils

The functional unit

- the portion of the myofibril between two successive Z lines is the functional unit of contraction, and is called a sarcomere
- at rest the edges of the thin filaments on either side of the thick filaments partly overlap the free ends of the thick filaments, leaving their central part clear
- that central part of the thick filament, not overlapped by thin filaments, is the H zone

Actin binding sites on the myosin head

Head, the globular part of the meromyosin

ATP binding sites, also on the head

Cross arm, the head with its short arm projecting outwards

Figure 17.C (a) An actin (thin) filament (b) Myosin monomer (Meromyosin)

17.5 Structure of the contractile proteins

The thin filament

- each actin or thin filament is made of two F, that is filamentous, actins helically wound to each other
- each F actin is a polymer of monomeric G, that is globular, actins
- two filaments of another protein, tropomyosin, also run close to the F actins along their whole length, and a complex protein, troponin, is distributed at regular intervals on the tropomyosin
- in the resting state a subunit of troponin masks the active binding sites for myosin on the actin filaments

The thick filament

- each myosin or thick filament is also a polymerised protein, built of many monomeric proteins called meromyosins
- each meromyosin has two important parts, a globular head with a short arm, the heavy meromyosin (HMM), and a tail, the light meromyosin (LMM)
- the HMM component, that is the head and short arm, projects outwards at a regular distance and angle from the surface of the polymerised myosin filament and is known as the cross arm
- the globular head is an active ATPase enzyme and carries binding sites for ATP and active sites for actin

17.6 Mechanism of muscle contraction

The sliding filament theory

- the mechanism of muscle contraction is best explained by the sliding filament theory
- it states that contraction of a muscle fibre takes place by the sliding of the thin filaments over the thick filaments

Myosin head binding to the exposed active site on actin

Cross bridge formed between thick and thin filament

Rotation of the head, pulling the actin filament along

Breaking of the cross bridge as a new ATP binds

Figure 17.D Stages in cross bridge formation, rotation of head and breaking of cross bridge

17.7 The contraction cycle

From neural signal to cross bridge

- contraction is initiated by a signal sent by the central nervous system (CNS) through a motor neuron, and a motor neuron together with the muscle fibres connected to it makes a motor unit
- the junction between a motor neuron and the sarcolemma of the muscle fibre is the neuromuscular junction or motor-end plate
- a neural signal reaching this junction releases the neurotransmitter acetyl choline, which generates an action potential in the sarcolemma, and this spreads through the fibre and causes the release of calcium ions into the sarcoplasm
- the rise in Ca++ level leads calcium to bind a subunit of troponin on the actin filaments and so unmask the active sites for myosin, and using the energy of ATP hydrolysis the myosin head binds the exposed active sites on actin to form a cross bridge

What the cross bridge does

- this pulls the attached actin filaments towards the centre of the A band
- the Z lines attached to these actins are pulled inwards as well, so the sarcomere shortens, which is contraction
- during shortening the I bands are reduced whereas the A bands keep their length
- the myosin then releases the ADP and P1 and goes back to its relaxed state, a new ATP binds and the cross bridge is broken, the ATP is hydrolysed again by the myosin head, and the cycle of cross bridge formation

Thin filaments sliding in over the thick filaments

I band shortening as contraction proceeds

H zone narrowing at the same time

A band keeping its length throughout

Figure 17.E Sliding-filament theory of muscle contraction (movement of the thin filaments and the relative size of the I band and H zones)

17.8 Relaxation, fatigue and fibre types

How the muscle relaxes

- the cycle of cross bridge formation and breakage repeats and causes further sliding
- the process continues until the Ca++ ions are pumped back to the sarcoplasmic cisternae, which masks the actin filaments again
- this returns the Z lines to their original position, that is relaxation, and the reaction time of the fibres can vary in different muscles
- repeated activation can build up lactic acid from the anaerobic breakdown of glycogen in the muscles, causing fatigue

Red fibres and white fibres

- muscle contains a red coloured oxygen storing pigment called myoglobin
- myoglobin content is high in some muscles, which gives them a reddish appearance, and these are the red fibres
- red fibres also hold plenty of mitochondria, which can use the large amount of oxygen stored in them for ATP production, so they can also be called aerobic muscles
- some muscles hold very little myoglobin and appear pale or whitish, and these white fibres have few mitochondria but a high amount of sarcoplasmic reticulum, and depend on an anaerobic process for energy

Frontal, parietal, temporal, occipital, sphenoid and ethmoid, the cranial bones

Nasal, lacrimal, zygomatic, maxilla and mandible, among the facial bones

Hyoid bone, at the base of the buccal cavity

Occipital condyles, where the skull meets the vertebral column

Figure 17.F Diagrammatic view of human skull

17.9 The skeletal system

Bones and cartilages

- the skeletal system consists of a framework of bones and a few cartilages, and it matters greatly in the movement the body shows
- bone and cartilage are specialised connective tissues, the first with a very hard matrix because of the calcium salts in it, the second with a slightly pliable matrix because of chondroitin salts
- in human beings the system is made up of 206 bones and a few cartilages, grouped into two principal divisions, the axial and the appendicular skeleton

The axial skeleton and the skull

- the axial skeleton comprises 80 bones distributed along the main axis of the body, namely the skull, the vertebral column, the sternum and the ribs
- the skull is composed of two sets of bones, cranial and facial, totalling 22
- the cranial bones are 8 in number and form the hard protective outer covering, the cranium, for the brain, while the facial region is made up of 14 skeletal elements forming the front part of the skull
- a single U-shaped bone, the hyoid, lies at the base of the buccal cavity, each middle ear holds three tiny bones, the malleus, incus and stapes, collectively the ear ossicles, and the skull articulates with the superior region of the vertebral column through two occipital condyles, making it a dicondylic skull

Cervical vertebra at the top of the column

Thoracic vertebra below it

Intervertebral disc between successive vertebrae

Lumbar vertebra, then the sacrum and the coccyx

Figure 17.G Vertebral column (right lateral view)

True ribs, the first seven pairs reaching the sternum

Vertebrochondral ribs, the 8th, 9th and 10th pairs

Floating ribs, the last two pairs, free in front

Sternum and thoracic vertebrae closing the cage

Figure 17.H Ribs and rib cage

17.10 The vertebral column and the rib cage

The column

- our vertebral column is formed by 26 serially arranged units called vertebrae and is dorsally placed, extending from the base of the skull and making the main framework of the trunk
- each vertebra has a central hollow portion, the neural canal, through which the spinal cord passes, and the first vertebra, the atlas, articulates with the occipital condyles
- the column is differentiated, starting from the skull, into cervical (7), thoracic (12), lumbar (5), sacral (1-fused) and coccygeal (1-fused) regions, the cervical vertebrae being seven in almost all mammals, human beings included
- it protects the spinal cord, supports the head and serves as the point of attachment for the ribs and for the musculature of the back

Sternum and ribs

- the sternum is a flat bone on the ventral midline of the thorax
- there are 12 pairs of ribs, each a thin flat bone connected dorsally to the vertebral column and ventrally to the sternum, with two articulation surfaces on its dorsal end, hence bicephalic
- the first seven pairs are the true ribs, attached dorsally to the thoracic vertebrae and connected ventrally to the sternum by hyaline cartilage
- the 8th, 9th and 10th pairs do not meet the sternum directly but join the seventh rib through hyaline cartilage and are called vertebrochondral or false ribs, while the last 2 pairs are not connected ventrally and are called floating ribs, and the thoracic vertebrae, ribs and sternum together form the rib cage

Clavicle and scapula, the two halves of one side

Spine of the scapula, projecting as the acromion

Glenoid cavity, receiving the head of the humerus

Humerus, radius and ulna of the upper arm

Figure 17.I Right pectoral girdle and upper arm. (frontal view)

Coxal bone, formed of ilium, ischium and pubis

Acetabulum, the cavity at their point of fusion

Femur, tibia and fibula of the lower limb

Patella, the cup shaped knee cap

Figure 17.J Right pelvic girdle and lower limb bones (frontal view)

17.11 The appendicular skeleton

Bones of the limbs

- the bones of the limbs, together with their girdles, make the appendicular skeleton, and each limb is made of 30 bones
- the bones of the hand, the fore limb, are the humerus, the radius and ulna, the carpals or wrist bones, 8 in number, the metacarpals or palm bones, 5 in number, and the phalanges or digits, 14 in number
- the bones of the leg, the hind limb, are the femur or thigh bone, the longest bone, the tibia and fibula, the tarsals or ankle bones, 7 in number, the metatarsals, 5 in number, and the phalanges or digits, 14 in number, and a cup shaped bone, the patella, covers the knee ventrally as the knee cap

17.12 The girdles

The pectoral girdle

- pectoral and pelvic girdle bones help the upper and the lower limbs articulate with the axial skeleton, and each girdle is formed of two halves
- each half of the pectoral girdle consists of a clavicle and a scapula
- the scapula is a large triangular flat bone in the dorsal part of the thorax between the second and the seventh ribs, its dorsal flat triangular body carrying a slightly elevated ridge, the spine, which projects as a flat expanded process, the acromion, with which the clavicle articulates
- below the acromion is a depression, the glenoid cavity, which articulates with the head of the humerus to form the shoulder joint, and each clavicle is a long slender bone with two curvatures, commonly called the collar bone

The pelvic girdle

- the pelvic girdle consists of two coxal bones
- each coxal bone is formed by the fusion of three bones, the ilium, the ischium and the pubis
- at the point of their fusion is a cavity, the acetabulum, to which the thigh bone articulates
- the two halves of the girdle meet ventrally to form the pubic symphysis, which contains fibrous cartilage

17.13 Joints

What a joint is

- joints are essential for all types of movement involving the bony parts of the body, and locomotory movements are no exception
- they are points of contact between bones, or between bones and cartilages
- the force generated by the muscles is used to carry out movement through joints, where the joint acts as a fulcrum, and the movability at these joints varies with different factors
- joints have been classified into three major structural forms, fibrous, cartilaginous and synovial

The three forms

- fibrous joints allow no movement, and are shown by the flat skull bones, which fuse end-to-end through dense fibrous connective tissues in the form of sutures to make the cranium
- in cartilaginous joints the bones are joined by cartilages, as between adjacent vertebrae in the vertebral column, and these permit limited movement
- synovial joints are marked by a fluid filled synovial cavity between the articulating surfaces of the two bones, an arragement that allows considerable movement and helps in locomotion and many other movements
- the ball and socket joint between humerus and pectoral girdle, the hinge joint of the knee, the pivot joint between atlas and axis, the gliding joint between the carpals and the saddle joint between carpal and metacarpal of the thumb are some examples

17.14 Disorders of the muscular and skeletal system

Muscle disorders

- myasthenia gravis is an auto immune disorder affecting the neuromuscular junction, leading to fatigue, weakening and paralysis of skeletal muscle
- muscular dystrophy is the progressive degeneration of skeletal muscle, mostly from a genetic disorder
- tetany is rapid spasms, that is wild contractions, in muscle, caused by low Ca++ in the body fluid

Bone and joint disorders

- arthritis is inflammation of joints
- osteoporosis is an age-related disorder marked by decreased bone mass and increased chances of fractures, a common cause being decreased levels of estrogen
- gout is inflammation of joints from the accumulation of uric acid crystals

17.15 Summary

Movement and locomotion

- movement is an essential feature of all living beings, and protoplasmic streaming, ciliary movements and movements of fins, limbs and wings are some of the forms animals show
- a voluntary movement that causes the animal to change its place is called locomotion

Muscles and their kinds

- animals move generally in search of food, shelter, a mate, a breeding ground, a better climate, or to protect themselves
- the cells of the human body show amoeboid, ciliary and muscular movements, and locomotion and many other movements need coordinated muscular activity
- three types of muscle are present, the skeletal, attached to skeletal elements, striated and voluntary, the visceral, in the inner walls of visceral organs, nonstriated and involuntary, and the cardiac, the muscle of the heart, striated, branched and involuntary
- muscles possess excitability, contractility, extensibility and elasticity

How a muscle contracts

- the muscle fibre is the anatomical unit of muscle, each fibre holding many parallelly arranged myofibrils, and each myofibril many serially arranged sarcomeres, the functional units
- each sarcomere has a central A band of thick myosin filaments and two half I bands of thin actin filaments on either side of it, marked by Z lines, and actin and myosin are polymerised proteins with contractility
- the active sites for myosin on the resting actin filament are masked by the protein troponin, and the myosin head contains ATPase and has ATP binding sites and active sites for actin
- a motor neuron carries a signal to the muscle fibre, generating an action potential that releases Ca++ from the sarcoplasmic reticulum, Ca++ activates actin, which binds the myosin head to form a cross bridge, the cross bridges pull the actin filaments so that they slide over the myosin filaments and the muscle contracts, and Ca++ is then returned to the reticulum, the actin is inactivated, the cross bridges break and the muscle relaxes

Fatigue, fibres and the skeleton

- repeated stimulation of muscles leads to fatigue, and muscles are classified as red and white fibres, based primarily on the amount of red coloured myoglobin pigment in them
- bones and cartilages constitute our skeletal system, which is divisible into the axial and the appendicular, the skull, vertebral column, ribs and sternum making the first and the limb bones and girdles the second
- three types of joint are formed between bones, or between bone and cartilage, the fibrous, the cartilaginous and the synovial, and synovial joints allow considerable movement and so matter greatly in locomotion

17.16 Exercises

From this chapter

1. Draw the diagram of a sarcomere of skeletal muscle showing different regions.
2. Define sliding filament theory of muscle contraction.
3. Describe the important steps in muscle contraction.

17.17 Exercises, continued

From this chapter, continued

1. Write true or false. If false change the statement so that it is true. (a) Actin is present in thin filament (b) H-zone of striated muscle fibre represents both thick and thin filaments. (c) Human skeleton has 206 bones. (d) There are 11 pairs of ribs in man. (e) Sternum is present on the ventral side of the body.
2. Write the difference between : (a) Actin and Myosin (b) Red and White muscles (c) Pectoral and Pelvic girdle
3. Match Column I with Column II : Column I (a) Smooth muscle (b) Tropomyosin (c) Red muscle (d) Skull , Column II (i) Myoglobin (ii) Thin filament (iii) Sutures (iv) Involuntary
4. What are the different types of movements exhibited by the cells of human body?
5. How do you distinguish between a skeletal muscle and a cardiac muscle?
6. Name the type of joint between the following: (a) atlas/axis (b) carpal/metacarpal of thumb (c) between phalanges (d) femur/acetabulum (e) between cranial bones (f) between pubic bones in the pelvic girdle
7. Fill in the blank spaces: (a) All mammals (except a few) have _____ cervical vertebra. (b) The number of phalanges in each limb of human is _____ (c) Thin filament of myofibril contains 2 'F' actins and two other proteins namely _____ and _____. (d) In a muscle fibre Ca++ is stored in _____ (e) _____ and _____ pairs of ribs are called floating ribs. (f) The human cranium is made of _____ bones.